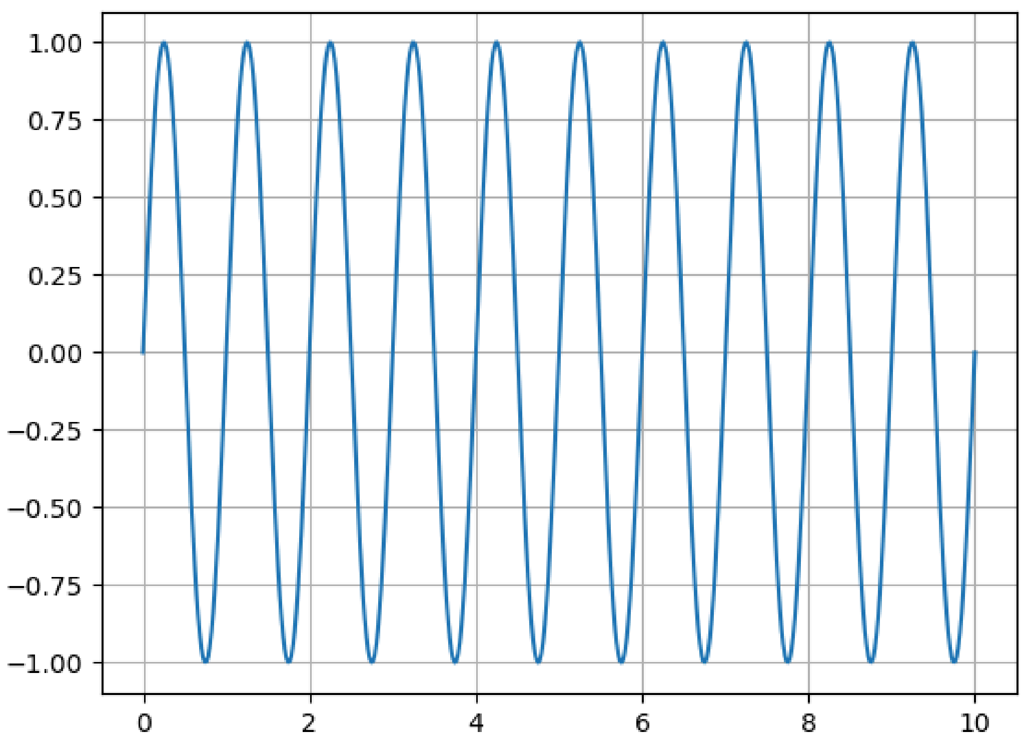
day 39

In [1]:
```
%store -r
```

In [2]:
```
plt.close()
```

In [3]:
```
plt.plot(x0, y0, 'C0')
plt.grid()
plt.show()
```

Figure



In [4]:
```
def dft(y):
    N = len(y)
    c = np.zeros(N//2+1, complex)
    for k in range(N//2+1):
        for n in range(N):
            c[k] += y[n]*np.exp(-2j*np.pi*k*n/N)
    return(c)
```

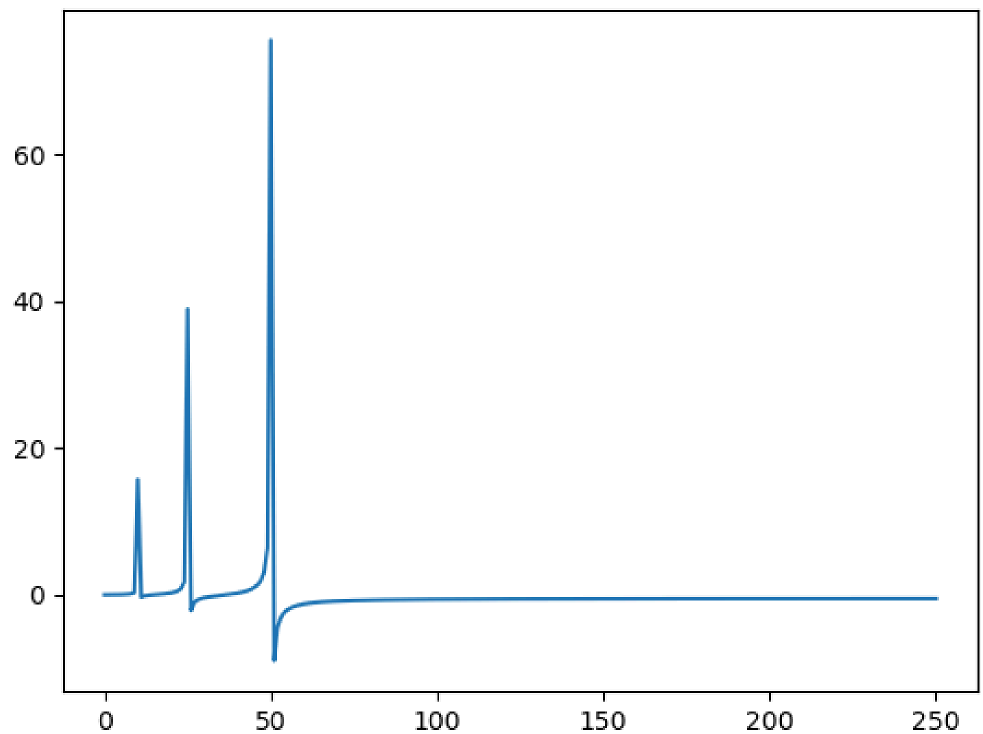
In [5]:
```
x0 = np.linspace(0,10, 500)
y0 = np.sin(2*np.pi*x0) + np.sin(5*np.pi*x0) + np.sin(10*np.pi*x0)
```

In [6]:
```
c = dft(y0)
```

In [7]:
```
fig0, ax0 = plt.subplots()
ax0.plot(c.real)
```

Out[7]: [<matplotlib.lines.Line2D at 0x7f0c97a32300>]

Figure

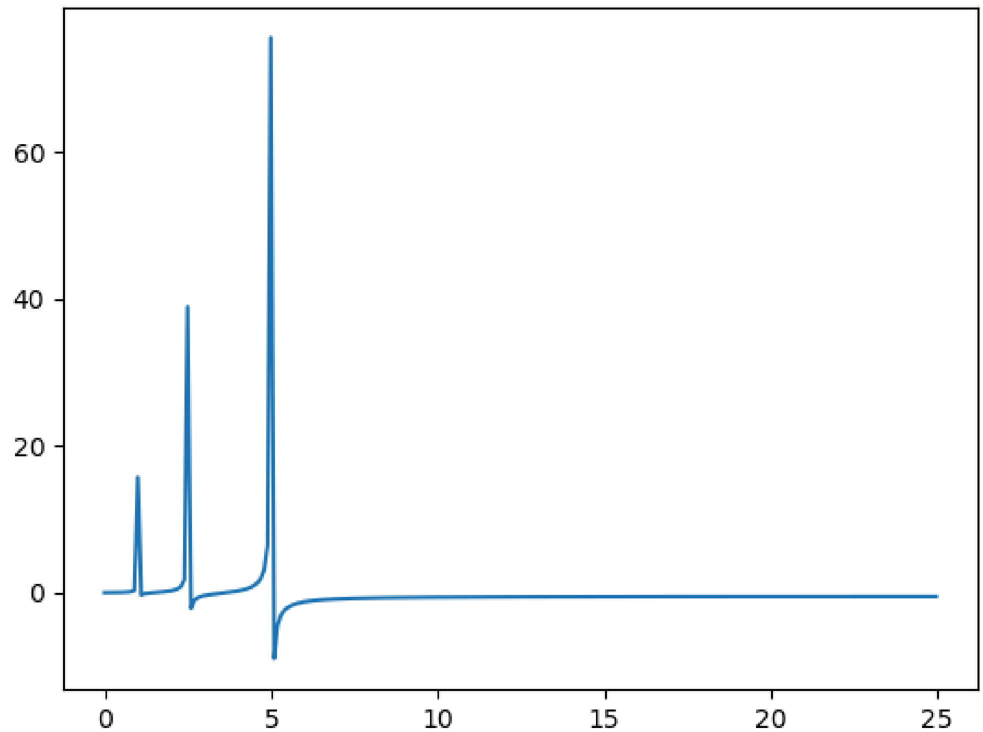


In [8]:
```
cfft = np.fft.rfft(y0)
dx = x0[1]-x0[0]
w = np.fft.rfftfreq(500, d=dx)
```

In [9]:
```
fig1, ax1 = plt.subplots()
ax1.plot(w, cfft.real)
```

Out[9]: [<matplotlib.lines.Line2D at 0x7f0c971f44a0>]

Figure

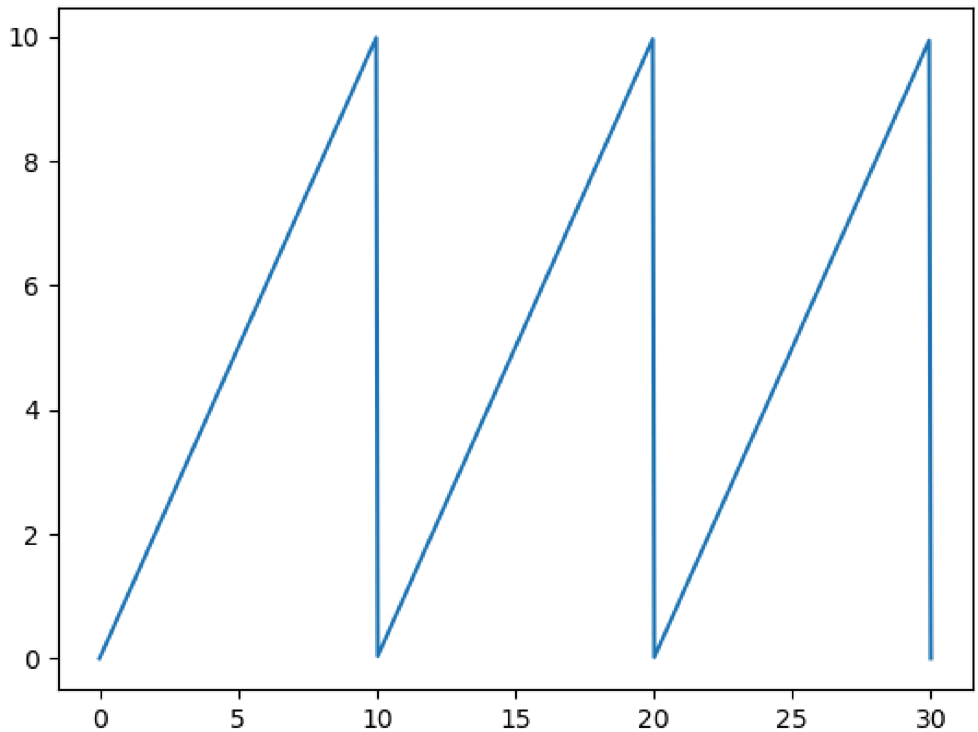


In [10]:
```
L = 10
x = np.linspace(0, 3*L, 500)
y = (x % L)
```

In [11]:
```
fig2, ax2 = plt.subplots()
ax2.plot(x, y)
```

Out[11]:
```
[<matplotlib.lines.Line2D at 0x7f0c9501c740>]
```

Figure

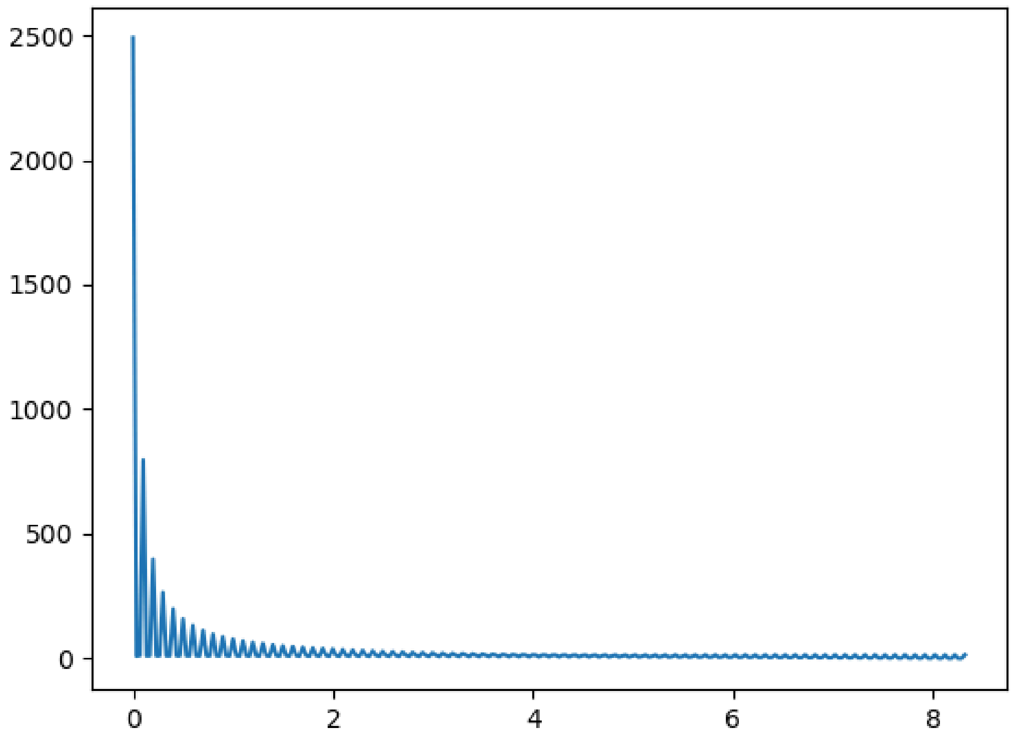


In [13]:
```
cfft = np.fft.rfft(y)
dx = x[1]-x[0]
w = np.fft.rfftfreq(500, d=dx)

fig4, ax4 = plt.subplots()
ax4.plot(w, np.sqrt(cfft*cfft.conjugate())) # here i am plotting the magnitu
```

Out[13]: [<matplotlib.lines.Line2D at 0x7f0c94f2fda0>]

Figure

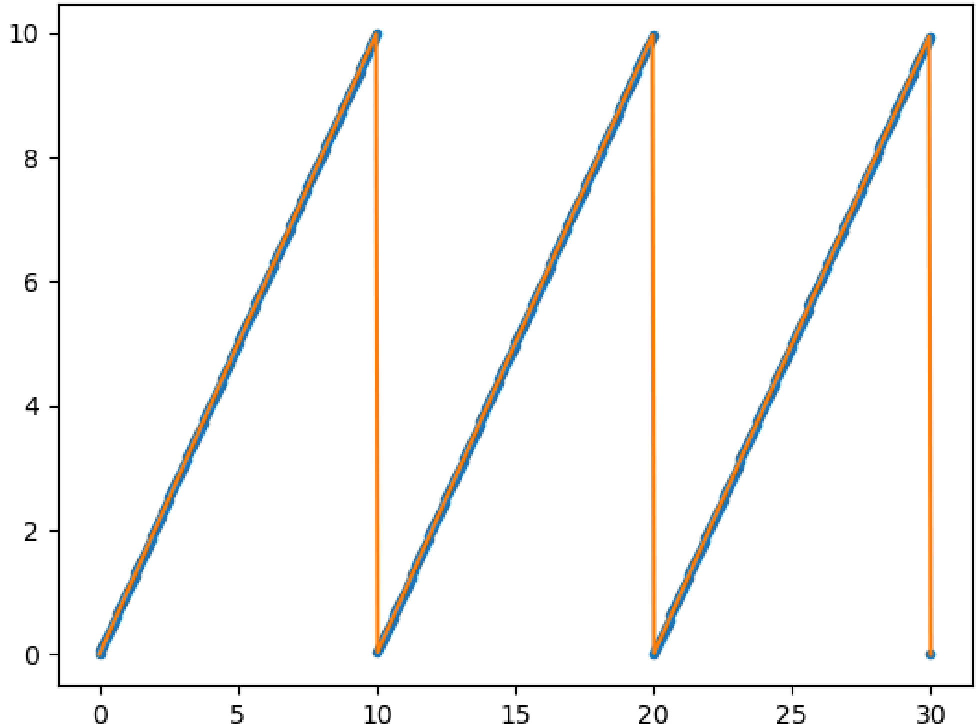


In [14]:
```
y5 = np.fft.irfft(cfft)
```

In [20]:
```
fig5, ax5 = plt.subplots()
ax5.plot(x, y5, '.')
ax5.plot(x, y)
```

Out[20]:
```
[<matplotlib.lines.Line2D at 0x7f0c94ef17f0>]
```

Figure



what if i cut out some of that fourier transform??

In [45]:
```
y6 = np.fft.irfft(cfft[0:251])
```

In [46]:
```
len(y6)
```

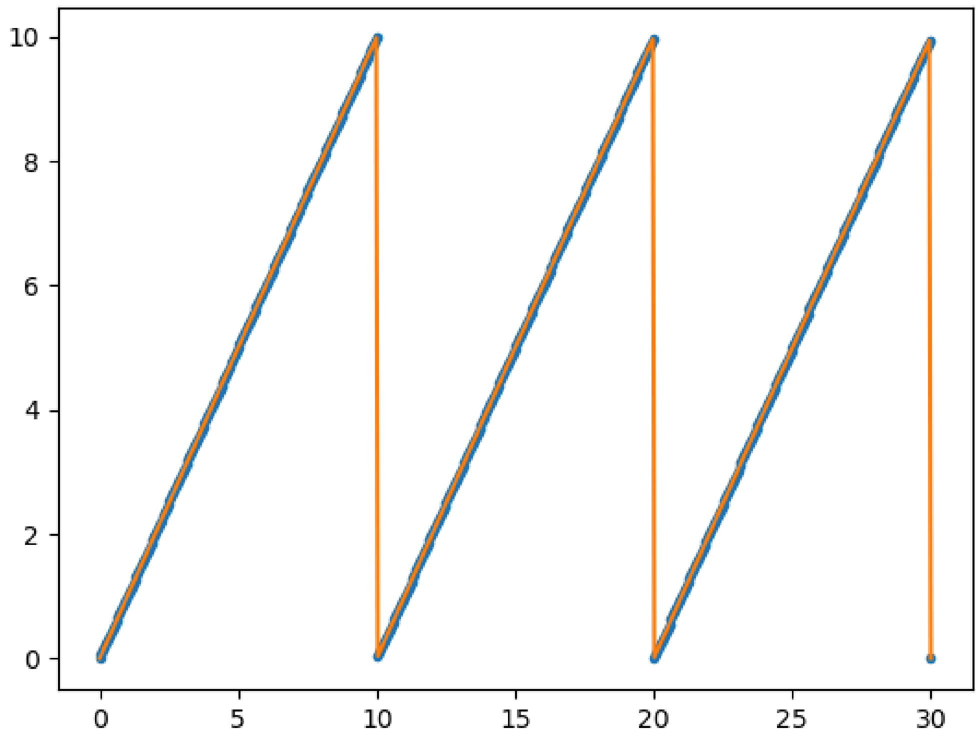
Out[46]:
```
500
```

In [47]:
```
fig6, ax6 = plt.subplots()
ax6.plot(x[:len(y6)], y6, '.')
ax6.plot(x, y)
```

```
/tmp/ipykernel_184494/1409771445.py:1: RuntimeWarning: More than 20 figures
have been opened. Figures created through the pyplot interface (`matplotlib.
pyplot.figure`) are retained until explicitly closed and may consume too muc
h memory. (To control this warning, see the rcParam `figure.max_open_warning
`). Consider using `matplotlib.pyplot.close()`.
  fig6, ax6 = plt.subplots()
```

Out[47]:
```
[<matplotlib.lines.Line2D at 0x7f0c949ade20>]
```

Figure



In [51]:
```
len(cfft)
```

Out[51]: 251

In [ ]: